Lecture 5: Input & Output Devices

Keyboard · Mouse · Touch Screen · Monitors (LCD, LED) · Printers (Inkjet, Laser)

Learning Outcomes

- Recall the Input Unit and Output Unit from Lecture 3, and explain why studying individual devices in detail matters for exams.
- Explain, step by step, how a keyboard converts a key press into a character on screen, including the role of a scan code.
- Explain how an optical mouse detects movement and converts it into cursor motion.
- Differentiate between resistive and capacitive touch screens, explaining how each detects a touch.
- Explain the working of LCD and LED monitors, and clarify the exact relationship between the two.
- Explain the working of inkjet and laser printers, including the five-step laser printing process.
- Compare all five devices using exam-ready tables, and answer “explain the working of…” and “differentiate between…” style questions confidently.
- Answer short, medium, and long-answer (up to 15 marks) examination questions on this topic with full explanations, diagrams, and examples.

1 Recap — From the Block Diagram to Individual Devices

In Lecture 3, the Input Unit and Output Unit each appeared as a single box in the block diagram — "accepts data from outside" and "presents the result to the user." This lecture opens up both of those boxes and studies five specific, commonly examined devices in real working detail: the keyboard, the mouse, the touch screen, the monitor (LCD and LED), and the printer (inkjet and laser).

Why This Matters for Exams

Questions on this topic are rarely just "name the device." They usually ask you to explain HOW a device works, step by step, or to differentiate between two closely related technologies (resistive vs capacitive, LCD vs LED, inkjet vs laser). This lecture is built specifically around answering exactly those question types.

2 The Keyboard

The keyboard is the primary input device used to enter text, numbers, and commands into a computer, arranged in the widely used QWERTY layout (named after the first six letters on the top row of letters).

A standard keyboard contains different types of keys according to their functions.

The main groups of keys are:

1. **Alphanumeric Keys:** These include alphabet keys **A–Z** and number keys **0–9**. They are mainly used for typing text and numbers.
2. **Function Keys:** These are **F1 to F12**, located at the top of the keyboard. They are used to perform specific functions. For example, **F1** is commonly used to open Help.
3. **Control Keys:** Keys such as **Ctrl, Alt, Shift, Esc, and Windows key** are control keys. They are often used alone or in combination with other keys to perform different operations.
4. **Navigation Keys:** These include **Arrow keys, Home, End, Page Up, and Page Down**. They are used to move the cursor and navigate through documents.
5. **Special/Editing Keys:** Keys such as **Enter, Spacebar, Backspace, Delete, Tab, and Caps Lock** are used for typing and editing operations.
6. **Numeric Keypad:** It is generally located on the right side of a full-sized keyboard. It contains **0–9** and mathematical operators such as *+, −, , / and is useful for entering numerical data quickly.

How a Keyboard Works — Step by Step

1. The user presses a key.

2. The keyboard's internal controller detects exactly which key was pressed (and later, released) and generates a unique scan code for that key.

3. This scan code is sent from the keyboard to the computer, usually over a USB connection.

4. The computer's keyboard driver/software translates the scan code into a character code (based on a standard such as ASCII or Unicode), taking the current keyboard language/layout into account.

5. The resulting character is passed to the active application and displayed on screen.

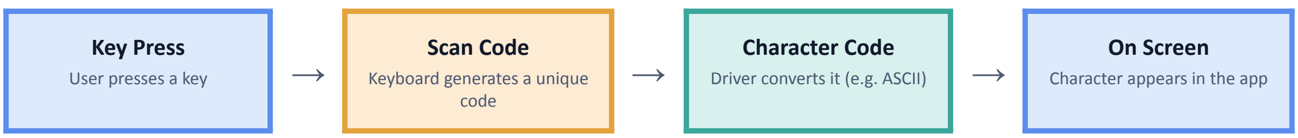


Fig 5.1 — How a key press becomes a character on screen.

Important Point for Exams

A scan code and a character code are NOT the same thing. The scan code simply identifies WHICH physical key was pressed; the character code is the meaningful letter, digit, or symbol that the computer displays, decided only after the scan code is translated. Examiners frequently test this exact distinction.

Type	How It Works
Membrane Keyboard	Keys press down onto a flexible rubber/plastic membrane sheet to complete a circuit; quieter and cheaper.
Mechanical Keyboard	Each key has its own individual spring-loaded switch; more durable, tactile, and often preferred by typists and gamers.

3 The Mouse

The mouse is a pointing input device used to control the position of the on-screen cursor and to select or interact with items via its buttons. . It helps the user to select, open, move, and perform different operations on items displayed on the screen..

The main parts of a mouse are:

1. **Left Button:** It is used to select items, open files and folders, and perform different commands. A single click selects an item, while a double-click usually opens it.
2. **Right Button:** It is used to display a **shortcut or context menu** containing additional options related to the selected item.
3. **Scroll Wheel:** It is located between the left and right buttons. It is used to scroll a page or document **up and down**.
4. **Sensor:** An optical or laser sensor is present at the bottom of the mouse. It detects the movement of the mouse and moves the pointer accordingly on the screen.

Common Mouse Operations:

- **Pointing:** Moving the mouse to place the pointer on an item.
- **Clicking:** Pressing and releasing the left mouse button once.
- **Double-clicking:** Pressing the left button twice quickly, usually to open a file or folder.
- **Right-clicking:** Pressing the right mouse button to display a shortcut menu.
- **Dragging and Dropping:** Holding the left button while moving the mouse to move an item from one place to another.
- **Scrolling:** Rotating the scroll wheel to move up or down a page.

How an Optical Mouse Works — Step by Step

1. A small LED (or laser) on the underside of the mouse shines light onto the surface below it.
2. A tiny image sensor captures thousands of images of that surface every second.
3. A built-in processor compares each new image with the previous one to work out the direction and distance the mouse has moved.
4. This movement is converted into a change in x and y coordinates and sent to the computer.
5. The computer moves the on-screen cursor by exactly that amount; separate signals are sent when a button is clicked.

LED Light Shines onto the surface	→	**Image Sensor** Captures rapid images	→	**Movement Calculated** Images compared for direction	→	**Cursor Moves** x, y sent to the computer

Fig 5.2 — How an optical mouse detects and reports movement.

Older mechanical mice used a rubber ball on the underside that rolled as the mouse moved, turning small internal rollers that measured motion — these are now obsolete, replaced entirely by optical and laser sensors, which are more accurate and have no moving parts to collect dirt.

4 Touch Screen — In Detail

A touch screen is an input (and output) device that allows a user to interact directly with what is displayed, by touching the screen itself rather than using a separate device like a mouse. The two most widely used touch technologies work in fundamentally different ways.

Working of a Touch Screen:
When a user touches the screen, the touch-sensitive surface detects the **position of the touch**. The device processes this information and performs the required action, such as opening an application, selecting an option, or typing a character.

Common Touch Screen Operations:

1. **Tap:** Touch the screen once to select or open an item.
2. **Double Tap:** Tap twice quickly to perform certain actions.
3. **Swipe:** Move a finger across the screen to change pages or scroll.
4. **Drag:** Touch and move an item from one position to another.
5. **Pinch:** Move two fingers closer together to **zoom out**.
6. **Spread:** Move two fingers apart to **zoom in**.

Uses of Touch Screens:
Touch screens are commonly used in **smartphones, tablets, ATMs, information kiosks, ticket machines, laptops, and self-service systems**.

Touch Screen Technologies	
Resistive • Detects physical PRESSURE • Works with any object	**Capacitive** • Detects electrical CHARGE • Needs a bare finger / conductive stylus

Fig 5.3 — The two main touch screen technologies.

Resistive Touch Screens

A resistive touch screen is built from two thin, flexible, electrically conductive layers, separated by a small gap. When you press down anywhere on the screen — with a finger, a stylus, a gloved hand, or any other object — the two layers touch at that exact point, completing an electrical circuit. The screen's controller measures the resulting change in electrical resistance to calculate the precise location touched.

Capacitive Touch Screens

A capacitive touch screen is coated with a material that stores an electrical charge across its surface, creating a uniform electrostatic field. A bare human finger (or a special conductive stylus) touching the screen draws away a tiny amount of this charge at that point. Sensors at the corners of the screen detect exactly where this change in capacitance occurred, and this is far more sensitive and precise than resistive touch, which is why capacitive screens can reliably support multi-touch gestures like pinch-to-zoom.

Resistive vs Capacitive — Full Comparison

Point	Resistive	Capacitive
What it detects	Physical pressure between two layers	Electrical charge from a conductive touch
Works with	Any object — finger, gloved hand, stylus, fingernail	Bare finger, or a special conductive stylus only
Multi-touch support	Poor / usually not supported	Excellent — supports gestures like pinch-to-zoom
Image clarity	Lower, due to extra flexible layers	Higher, better light transmission
Cost	Cheaper	More expensive
Typical devices	Older PDAs, ATMs, industrial equipment	Smartphones, tablets, modern laptops

Real-Life Example

You can operate an older bank ATM's resistive touch screen with a gloved hand in winter, but the same glove will not work on a modern smartphone's capacitive screen — because a glove is not electrically conductive, the capacitive screen simply cannot detect it.

5 Output Devices — A Quick Bridge

Having looked at three input devices in detail, we now turn to output — the devices that convert the CPU's binary result back into something a human being can see or hold. This lecture covers two of the most commonly examined output devices in detail: the monitor and the printer.

6 Monitors — LCD vs LED

A monitor (also called a VDU, or Visual Display Unit) presents output as a soft copy — a temporary, on-screen image that disappears once the device is switched off. Almost all modern monitors are flat-panel displays, built around liquid crystal technology.

LCD — Liquid Crystal Display

LCD stands for **Liquid Crystal Display**. It uses **liquid crystals** to produce images on the screen. These crystals do not produce light themselves, so a separate **backlight** is required.

Features of LCD Monitor:

- It has a **thin and lightweight** design.
- It consumes less electricity than older CRT monitors.
- It produces clear and sharp images.
- Traditional LCD monitors commonly use **CCFL (Cold Cathode Fluorescent Lamp)** for backlighting.
- It occupies less space.

LED — Light Emitting Diode Display

LED stands for **Light Emitting Diode**. An LED monitor is actually a type of **LCD monitor that uses LEDs for backlighting** instead of CCFL lamps.

Features of LED Monitor:

- It is **thin and lightweight**.
- It consumes less power.
- It provides good **brightness and contrast**.
- It generally produces better picture quality than older CCFL-backlit LCD monitors.
- It is more energy-efficient.

Important Point for Exams — A Favourite Trick Question

Do NOT describe LCD and LED as two completely separate display technologies. An LED monitor IS an LCD monitor — the only difference is the type of backlight used (LED instead of CCFL). This distinction is one of the most frequently misunderstood points in this topic, and examiners often specifically test it.

Point	Traditional LCD (CCFL backlight)	LED-backlit LCD
Backlight source	CCFL (Cold Cathode Fluorescent Lamp) tubes	An array of LEDs
Thickness	Thicker	Thinner
Power consumption	Higher	Lower
Contrast	Lower	Better, especially with local dimming
Lifespan	Shorter	Longer

7 Printers — Inkjet vs Laser

A printer converts the CPU's binary result into a hard copy — a permanent, physical output on paper. The two most common printer technologies work in very different ways.

Inkjet Printers

An inkjet printer forms text and images by spraying extremely tiny droplets of liquid ink, through a set of microscopic nozzles, directly onto the paper in the required pattern. Most inkjet printers use separate cartridges for cyan, magenta, yellow, and black ink (CMYK), blending them to produce a full range of colours — which is why inkjet printers are generally preferred for photographs and colour-rich documents.

Features of Inkjet Printer:

- It can print both **black-and-white and colour** documents.
- It provides good-quality **photo and image printing**.
- It is generally cheaper to purchase.
- It uses **ink cartridges or ink tanks**.
- It is commonly used in **homes and small offices**.
- Its printing speed is generally lower than that of a laser printer.

Laser Printers

A laser printer works very differently, using a laser beam, static electricity, dry toner powder, and heat rather than liquid ink.

Features of Laser Printer:

- It provides **fast and high-quality printing**.
- It is suitable for printing a large number of documents.
- It uses **toner** instead of liquid ink.
- It produces sharp and clear text.
- It is commonly used in **offices, schools, colleges, and businesses**.
- Its initial cost is generally higher than an inkjet printer.

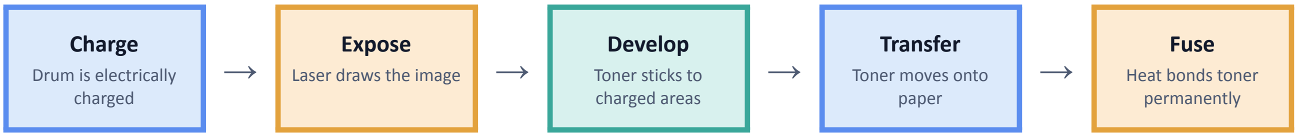


Fig 5.4 — The five-step laser printing process: Charge → Expose → Develop → Transfer → Fuse.

1. Charge: a rotating drum inside the printer is given a uniform electrical charge.
2. Expose: a laser beam scans across the drum, selectively discharging it in the exact pattern of the text or image to be printed.
3. Develop: fine toner powder, which is itself electrically charged, sticks only to the areas of the drum that were discharged by the laser.
4. Transfer: the drum rolls over the paper, and the toner pattern is transferred from the drum onto the paper's surface.
5. Fuse: the paper passes through a fuser unit, which applies heat and pressure to permanently melt and bond the toner onto the paper.

Inkjet vs Laser — Full Comparison

Point	Inkjet	Laser
Printing medium	Liquid ink, sprayed through nozzles	Dry toner powder, fused with heat
Speed	Slower	Much faster, especially for text
Best suited for	Photos and colour-rich documents	Bulk text printing, offices
Cost per page	Higher over time (ink is costly)	Lower over time, especially at high volume
Printer hardware cost	Cheaper upfront	More expensive upfront
Print quality on plain text	Good, can smudge if wet	Sharp, smudge-resistant

Point	Inkjet	Laser
Real-Life Example A college office printing 2,000 pages of an exam question bank, almost entirely plain black text, would choose a laser printer for its speed and low cost per page. A student printing a handful of colour photographs for a project would more likely choose an inkjet printer for its superior colour and photo quality.		

8 Input & Output Devices — Full Recap

Input Devices

Device	What It Converts	Key Working Principle
Keyboard	Key press → character	Scan code, translated into a character code
Mouse	Hand movement → cursor movement	LED + image sensor detects motion, sent as x/y coordinates
Resistive Touch Screen	Pressure → location	Two flexible layers touch, resistance change is measured
Capacitive Touch Screen	Conductive touch → location	Electrostatic field disrupted, capacitance change is measured

Output Devices

Device	What It Produces	Key Working Principle
LCD Monitor	Soft copy, on-screen image	Liquid crystals control light from a CCFL backlight
LED Monitor	Soft copy, on-screen image	An LCD monitor using LEDs instead of CCFL for backlighting
Inkjet Printer	Hard copy, on paper	Liquid ink sprayed through nozzles
Laser Printer	Hard copy, on paper	Toner powder fused onto paper using a charged drum and heat

9 Putting It All Together — A Worked Example

Consider a student completing and submitting an assignment using a touchscreen laptop:

Step	Device Involved	What Happens
1	Keyboard	The student types the assignment; each key press is converted to a scan code, then a character.
2	Touch Screen (Capacitive)	The student taps and scrolls the touchscreen laptop display with a bare finger to navigate menus.
3	Mouse / Trackpad	The student uses a mouse to precisely select and format a paragraph of text.
4	LED Monitor	Throughout, the student views their work on the laptop's LED-backlit LCD screen — a soft copy.
5	Laser Printer	Once finished, the student prints the assignment at a college printing shop using a laser printer, for a fast, sharp, low-cost hard copy.

This single everyday task uses every device studied in this lecture — three input devices and two output devices — exactly the range this lecture set out to explain in exam-ready detail.

10 Fill in the Blanks

- 1. When a key is pressed, the keyboard generates a unique ______, which is sent to the computer.
- 2. A modern optical mouse uses a(n) ______ and an image sensor to detect movement across a surface.
- 3. A ______ touch screen works by detecting physical pressure between two flexible layers.
- 4. A ______ touch screen works by sensing the electrical charge of a bare finger.
- 5. ______ touch screens support true multi-touch gestures like pinch-to-zoom.
- 6. An LED monitor is actually a type of ______ display that uses LEDs for backlighting.
- 7. In a laser printer, the toner is a fine ______, not a liquid.
- 8. In a laser printer, the ______ unit uses heat and pressure to permanently bond toner to the paper.
- 9. An inkjet printer forms an image by spraying tiny droplets of ______ onto paper.
- 10. The step in laser printing where the laser beam draws the image onto the drum is called ______.

Answers

1. Scan code | 2. LED (light source) | 3. Resistive | 4. Capacitive | 5. Capacitive | 6. LCD (Liquid Crystal Display) | 7. Powder | 8. Fuser | 9. Liquid ink | 10. Exposing (Exposure)

11 True or False

- 1. A resistive touch screen can be operated with a gloved hand or any stylus.
- 2. A capacitive touch screen works by detecting physical pressure.
- 3. LED monitors are a completely different, unrelated technology from LCD monitors.

- 4. A keyboard sends a scan code, which the computer then converts into a character code.
- 5. Laser printers use liquid ink cartridges.
- 6. Inkjet printers are generally better suited to bulk, high-volume text printing than laser printers.
- 7. An optical mouse uses an LED and an image sensor to track movement.
- 8. Capacitive touch screens generally provide better image clarity than resistive touch screens.

Answers with Explanation

1. True. Resistive screens respond to physical pressure from any object, including a gloved hand or a plain, non-conductive stylus.

2. False. It is the reverse: a capacitive screen senses electrical charge from a conductive touch, such as a bare finger, not physical pressure.

3. False. An LED monitor is a type of LCD monitor that uses LEDs for backlighting instead of CCFL tubes — it is not an unrelated technology.

4. True. That is exactly the two-step translation: the scan code identifies the key first, and the character code is decided afterwards.

5. False. Laser printers use dry toner powder, not liquid ink; liquid ink is used by inkjet printers.

6. False. It is the reverse: laser printers are faster and cheaper per page for bulk text printing.

7. True. That is the standard mechanism of a modern optical mouse.

8. True. Capacitive screens use fewer and better light-transmitting layers, giving clearer images than resistive screens.

12 Short Answer Questions (2–3 Marks)

- Explain, in brief, how a keyboard converts a key press into a character on screen. (3 marks)
- What is a scan code? (2 marks)
- Explain how an optical mouse detects movement. (3 marks)
- Differentiate between a resistive and a capacitive touch screen. (3 marks)
- Why do capacitive touch screens support multi-touch, while resistive touch screens generally do not? (2 marks)
- Is an LED monitor a completely different technology from an LCD monitor? Explain. (3 marks)
- Differentiate between an inkjet printer and a laser printer. (3 marks)
- Name the five steps in the laser printing process, in order. (2 marks)
- Give one real-life example each of a device suited to a resistive touch screen and one suited to a capacitive touch screen. (2 marks)

13 Long Answer Model Answers (5, 10 & 15 Marks)

The following are fully written model answers. Use these as a template for structure, depth, and the level of explanation expected in your own exam answer — do not simply memorise them word for word; practise writing them in your own words using the same structure.

Q1 (5 Marks): Explain how a keyboard and a mouse convert physical human action into computer input.

A keyboard converts a physical key press into computer input through a two-step translation. First, when a key is pressed, the keyboard's internal controller generates a unique scan code identifying exactly which key was used, and sends this scan code to the computer. Second, the computer's keyboard driver translates that scan code into a meaningful character code, such as ASCII, based on the current keyboard layout, and this character is then displayed in the active application.

A mouse converts hand movement into computer input using light and an image sensor. An LED on the underside of the mouse shines onto the surface below, and a tiny image sensor captures thousands of images of that surface every second. A built-in processor compares consecutive images to calculate the exact direction and distance moved, converts this into a change in x and y coordinates, and sends it to the computer, which moves the on-screen cursor by that same amount.

Q2 (10 Marks): Differentiate between resistive and capacitive touch screens, explaining how each works, with examples.

Touch Screen Technologies	
Resistive • Detects physical PRESSURE • Works with any object	**Capacitive** • Detects electrical CHARGE • Needs a bare finger / conductive stylus

(The two touch screen technologies, as in Fig 5.3 of these notes.)

A resistive touch screen is built from two thin, flexible, conductive layers separated by a small gap. Pressing anywhere on the screen — with a finger, a stylus, or even a gloved hand — pushes the two layers together at that point, completing an electrical circuit; the screen's controller calculates the touch location from the resulting change in electrical resistance. Because it responds to pressure from any object, a resistive screen works even with a gloved hand, but it generally cannot support true multi-touch gestures and produces a slightly less clear image due to its extra flexible layers.

A capacitive touch screen is coated with a material holding a uniform electrical charge across its surface. A bare human finger, being electrically conductive, draws away a tiny amount of this charge at the point of contact; sensors around the screen detect precisely where this change in capacitance occurred. Because this method depends on conductivity rather than pressure, a capacitive screen does not respond to a plain gloved hand or a non-conductive stylus, but it is far more sensitive and precise, supporting multi-touch gestures such as pinch-to-zoom, and gives a clearer image.

Point	Resistive	Capacitive
Detects	Physical pressure	Electrical charge
Works with	Any object, including gloves	Bare finger or conductive stylus only
Multi-touch	Poor	Excellent
Example device	Older ATM screens	Smartphones

In conclusion, the fundamental difference lies in what each screen actually measures — pressure versus charge — and this single difference explains every other practical distinction between them, from glove compatibility to multi-touch support to image clarity.

Q3 (10 Marks): Explain the working of LCD and LED monitors, and clarify the relationship between the two.

An LCD (Liquid Crystal Display) monitor is built from a layer of liquid crystals sandwiched between polarising filters and glass. The liquid crystals do not produce light themselves; instead, a backlight behind the whole panel shines light through it, and an electrical voltage applied to each cell twists its liquid crystals to control exactly how much of that backlight is allowed to pass through at that point. Colour filters over each cell then produce red, green, and blue sub-pixels, which combine to create a full-colour image. Traditional LCD monitors used CCFL (Cold Cathode Fluorescent Lamp) tubes to provide this backlight.

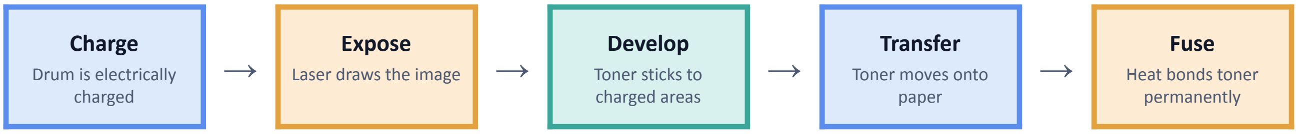
An LED monitor is not a separate display technology — it is simply an LCD monitor that uses an array of LEDs for its backlight instead of CCFL tubes. Because LEDs are smaller, more energy-efficient, and can even be dimmed individually in different zones of the screen, LED-backlit LCD monitors are thinner, consume less power, and can achieve noticeably better contrast (especially with a feature called local dimming) than older CCFL-backlit LCD monitors.

Point	Traditional LCD (CCFL backlight)	LED-backlit LCD
Backlight source	CCFL tubes	An array of LEDs
Thickness	Thicker	Thinner
Power consumption	Higher	Lower
Contrast	Lower	Better, especially with local dimming

In conclusion, every LED monitor is fundamentally an LCD monitor at its core — the term “LED” refers only to the type of backlight used, not to a different way of forming the image itself. Confusing the two as separate, unrelated technologies is one of the most common mistakes made in this topic.

Q4 (15 Marks): Explain the working of a laser printer in detail with a diagram, and compare it with an inkjet printer. Explain which one you would choose for different real-world scenarios.

A laser printer produces a hard copy using a laser beam, static electricity, dry toner powder, and heat, through a fixed five-step process.



(The five-step laser printing process, as in Fig 5.4 of these notes.)

First, in the Charge step, a rotating drum inside the printer is given a uniform electrical charge. Second, in the Expose step, a laser beam scans across the drum, selectively discharging it in exactly the pattern of the text or image to be printed. Third, in the Develop step, electrically charged toner powder sticks only to the areas of the drum that the laser discharged. Fourth, in the Transfer step, the drum rolls over the paper and deposits this toner pattern onto its surface. Finally, in the Fuse step, the paper passes through a fuser unit that applies heat and pressure, permanently melting and bonding the toner onto the paper.

An inkjet printer works very differently: it sprays extremely tiny droplets of liquid ink through microscopic nozzles directly onto the paper, typically blending cyan, magenta, yellow, and black ink to produce colour. There is no drum, no static charge, and no heat-fusing step.

Point	Inkjet	Laser
Printing medium	Liquid ink	Dry toner powder
Speed	Slower	Much faster
Best suited for	Photos, colour documents	Bulk text, offices
Cost per page at volume	Higher	Lower

Choosing between them depends on the task. A college office printing thousands of pages of plain-text exam papers would choose a laser printer, since its speed and low cost per page make it far more economical at high volume. A student printing a handful of colour photographs for a personal project would more likely choose an inkjet printer, since its liquid-ink blending produces richer, smoother colour gradients better suited to photographic images.

In conclusion, the two technologies are built around fundamentally different physical processes — electrostatics and heat-fused powder for laser, versus sprayed liquid droplets for inkjet — and this difference in mechanism directly explains why each is better suited to a different real-world printing need.

14 Exam Tips

- Memory trick for the keyboard: “Press → Scan code → Character code → Screen.” Never skip the scan-code step when explaining how a keyboard works.
- Memory trick for touch screens: Resistive = PRESSURE (works with anything); Capacitive = CHARGE (needs a bare finger or special stylus).
- Memory trick for the laser printing process: “Charge, Expose, Develop, Transfer, Fuse” — the order is always tested; learn it as a fixed five-word sequence.
- Common Mistake: Do not describe LED monitors as a completely different technology from LCD monitors — an LED monitor is an LCD monitor with an LED backlight. This is one of the most frequently asked “clarify the relationship” questions.
- Common Mistake: Do not say laser printers use “ink” — they use dry toner powder. Only inkjet printers use liquid ink.
- For “differentiate” questions (resistive vs capacitive, LCD vs LED, inkjet vs laser), always answer using a table — it is faster to write and easier for the examiner to award marks clearly.
- For 10 and 15 mark “explain the working of…” questions, always draw the relevant diagram (Fig 5.1–5.4) and structure your answer as a numbered, step-by-step sequence — exactly like the model answers in Section 13.

Lecture 5 — Input & Output Devices — building directly on Lecture 3's Block Diagram of a Computer.